Project Proposal: PermitPulse

Introduction

Understanding what drives construction costs is important for urban planners, developers, and investors in a rapidly growing city like Toronto. The cost of housing in the city has reached record highs, with the average home price exceeding $1.1 million, even as the population continues to climb, making affordability a critical challenge for residents and policymakers alike (CP24, 2025). Higher development fees, regulatory delays, and conditions at the site have been seen as key contributors of such escalating costs. These insights are crucial for making informed decisions about where and how to invest in future development.

This project explores the following research question: **What are the key factors that influence the estimated construction cost of building permits in Toronto from 2017 to 2024?** Specifically, we aim to determine how project characteristics such as building type, structure type, number of new residential units, location, and floor area contribute to variations in estimated construction cost. Our analysis is based on over 100,000 cleared building permit records provided by the City of Toronto's open data portal.

We apply multiple linear regression, a statistical method appropriate to calculate the measures of association between a single continuous outcome variable (EST_CONST_COST) and several predictor variables. The technique accommodates both categorical predictors (e.g., permit type, structure type, ward) and continuous predictors (e.g., floor area, dwelling units), where all the factors are simultaneously estimated after controlling for others. Linear regression is appropriate in this context because the outcome variable is continuous, the method produces interpretable coefficients that estimate the effect of each predictor, and key model assumptions such as linearity, homoscedasticity, and normality of residuals can be evaluated using diagnostic plots.

This approach is also supported by past research. In a study examining cost estimation models for high-rise residential complexes, predictor variables of location and gross floor area were found to be robust predictors of construction cost, with log-linear regression improving both model interpretability and fit (Elhag et al., 2005). By aligning our analysis with both current housing affordability concerns and established modeling practices, this project has the potential to produce results that guide smarter, data-driven construction planning in Toronto.

Data Description

The dataset used in this analysis is the “Building Permits – Cleared Permits” dataset from the City of Toronto’s Open Data portal. It contains records of completed or closed building permits across Toronto. A building permit is a legal authorization, issued under the Building Code Act, required for construction, demolition, or major structural modifications.

Data is collected through the City’s permit application and inspection system. Applicants submit details such as structure type, scope, and estimated cost, and City staff update records during review and inspection. Completion is recorded when permits are officially closed.

The **response variable** in our model is **Estimated Construction Cost (EST_CONST_COST)**, reflecting the proposed cost of construction declared at the time of application and approved by the City. This variable is **continuous and numeric**, making it appropriate for linear regression. To improve data quality, we removed observations with zero or missing costs (around 20,000 rows). The variable exhibits **right-skewed distribution**, indicating the presence of large-value outliers that represent large-scale developments.

We selected the following **predictors** based on domain relevance, variation, and statistical suitability:

Table 1. Predictor summary table (with Predictor, Type, Description)

Predictor	Type	Description / Summary
Permit Type	Categorical	Captures functional differences (e.g., Designated Structure, Partial). Set baseline to "Residential Building Permit" for more meaningful interpretation.

Structure Type	Categorical	Describes physical form of buildings (e.g., Detached, Townhouse, Apartment).
Ward	Categorical	Geographical location via Toronto's ward-based grid system
Dwelling Units Created	Numerical	Reflects intensity of residential development. Right-skewed; most are low, some exceed 100 units.
Residential (sq m)	Numerical	Residential floor area. Larger values generally relate to larger projects and higher costs.

Table 2. Numeric summary statistics

EST_CONST_COST	DWELLING_UNITS_CREATED	RESIDENTIAL
Min. : 1	Min. : 0.000	Min. : 0.00
1st Qu.: 25000	1st Qu.: 0.000	1st Qu.: 0.00
Median : 100000	Median : 0.000	Median : 0.00
Mean : 623597	Mean : 1.147	Mean : 129.08
3rd Qu.: 300000	3rd Qu.: 0.000	3rd Qu.: 37.25
Max. :515000000	Max. :1531.000	Max. :125872.50

Table 3. Top 5 categories of categorical predictors

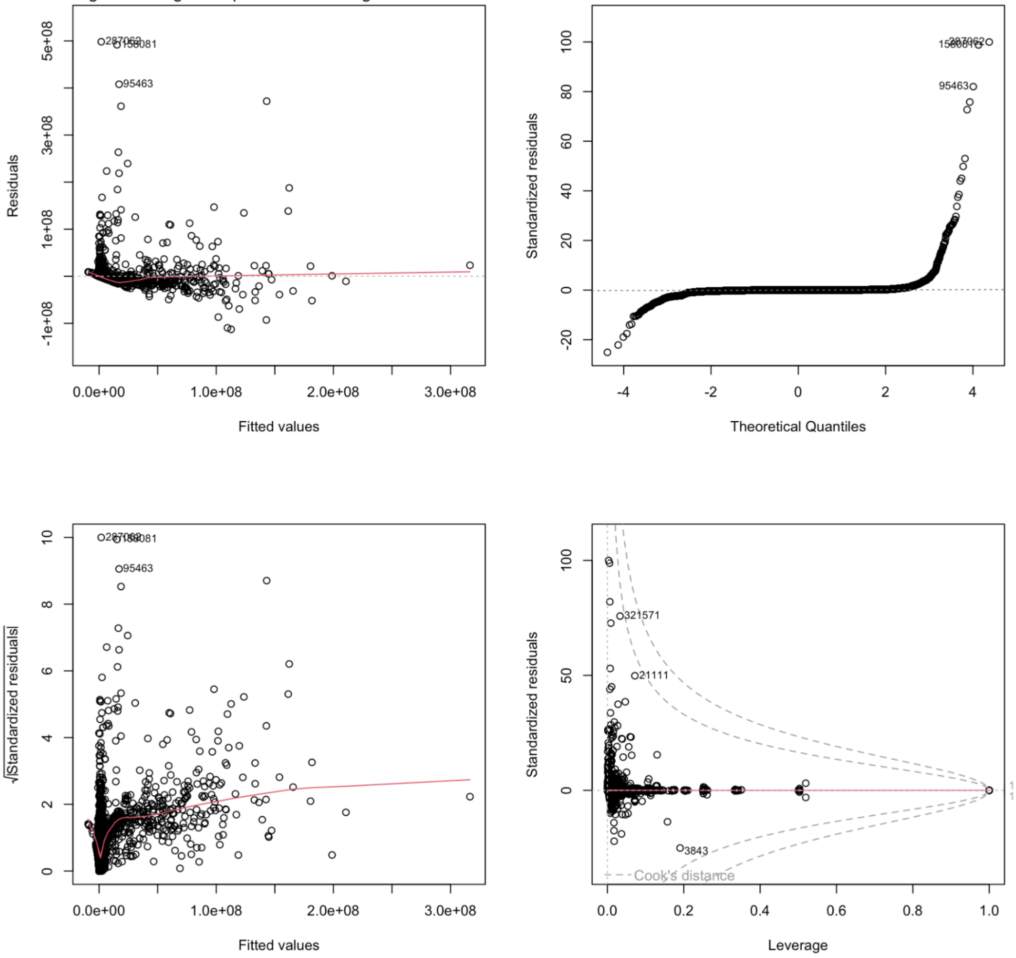
Permit_Type	Permit_Count	Structure_Type	Structure_Count	Ward	Ward_Count
Small Residential Projects	42134	SFD - Detached	32453	S1034	1390
Building Additions/Alterations	26461	SFD - Semi-Detached	9295	N1824	829
New Houses	11778	Office	8664	N0823	745
New Building	517	SFD - Townhouse	3903	S1128	745
Demolition Folder (DM)	513	2 Unit - Detached	3697	S1337	706

These variables capture spatial, structural, and regulatory influences on cost. **Table 2** shows right-skewed distributions in numeric predictors, with high-cost outliers. **Table 3** displays top categories; small residential and detached projects are most common, with some wards seeing more activity. Together, these summaries highlight the relevance and diversity of the predictors used in our model.

Preliminary Results

Figure 1. Regression diagnostic plots for the multiple linear regression model predicting construction costs.

Figure 1. Diagnostic plots for linear regression model.



We fitted a multiple linear regression model to examine how project size, type, and location predict estimated construction cost of building permits cleared in Toronto from 2017 to 2024. The response variable was EST_CONST_COST, and predictors included PERMIT_TYPE, STRUCTURE_TYPE, WARD_GRID, DWELLING_UNITS_CREATED, and RESIDENTIAL. After cleaning missing values, the dataset contained approximately 116,000 observations.

Residential Building Permit was chosen as the baseline category for PERMIT_TYPE because it represents a large and meaningful subset of permits with construction costs. The default baseline (Building Historical data – Converted) consisted mainly of administrative records and cancelled permits with zero cost, making comparisons less interpretable. Using Residential Building Permit as the reference allows all other permit type coefficients to be compared to a relevant construction category.

The results show that permit type and location are strong predictors of construction cost. For instance, New Building permits are associated with an increase of approximately **$8.2 million ($p < 0.001$)**, while Demolition Folder (DM) permits show a reduction of $267,000 ($p = 0.0091$). Small Residential Projects also have lower costs, with a decrease of $335,000 ($p = 0.043$). Many WARD_GRID levels were significant ($p < 0.01$), with estimated cost increases ranging from $6.5 million to $8.8 million compared to the baseline ward.

The model explains **45.3% of variability** in construction costs ($R^2 = 0.453$; $Adjusted\ R^2 = 0.449$), and the **F-statistic is 123.3** with a **p-value < 0**, indicating the model is statistically significant. As shown in Figure 1, the Residuals vs Fitted plot shows a curved pattern and increasing spread, suggesting non-linearity and heteroscedasticity. The Normal Q-Q plot shows deviations at the tails, indicating non-normal residuals. The Scale-Location plot reinforces heteroscedasticity. The Residuals vs Leverage plot highlights several high-leverage points with large Cook's distances, suggesting certain observations strongly influence the model. This suggests that log-transforming EST_CONST_COST in future analyses may improve model fit.

Bibliography

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